Question	Answer	Marks	Partial Marks	Guidance
10(b)	Rotation 90° anticlockwise oe [centre] (3, –1)	**3**	**B1** for each	More than one transformation scores zero Not turn Accept 90, 270 clockwise, – 270, 90 counterclockwise, 90 a/c Condone omission of brackets in coords, accept '<u>position</u> vector' $\begin{pmatrix} 3 \\ -1 \end{pmatrix}$ but not vector $\begin{pmatrix} 3 \\ -1 \end{pmatrix}$ If the centre has been described as a vector (word or bracket notation) then do not treat as additional transformation
11(a)	1842.59	**2**	**M1** for $1500 \times \left(1+\frac{4.2}{100}\right)^5$ oe	Mark at most accurate For 2 marks accept 1840, 1842, 1843, 1842.5, 1842.6, 1842.59… M1 may be done in stages and method can be implied by values given to 3 sf or better from previous stage[s] $1500 \times (1 + 4.2\%)^5$ is not enough for M1 Method spoiled if e.g. $1500 + 1500 \times \left(1+\frac{4.2}{100}\right)^5$ but allow M1 if $1500 \times \left(1+\frac{4.2}{100}\right)^5 - 1500$ M1 implied by answer 343, 342.5, 342.6, 342.59…

Question	Answer	Marks	Partial Marks	Guidance
11(b)	6.5[0…]	**3**	**M2** for $\sqrt[11]{2}$ or $2^{\frac{1}{11}}$ or **M1** for $[k\times]y^{11} = 2[\times k]$ oe	Mark at most accurate M2 implied by 1.065[0…] or 106.5[0…] or 0.065[0…] or 1.07 nfww or 107nfww M2 not for e.g. $\sqrt[11]{\frac{2y}{y}}$ e.g. uses a value or letter for the investment $1500\,y^{11} = 3000$
12	267 or 267.0 to 267.1	**2**	**M1** for $\pi\times12\times5+\pi\times5^2$	Mark at most accurate Accept 85 π as final answer Allow 3.14 or $\frac{22}{7}$ for π for M1 Answer in the wider range 266.9 to 267.1… not scoring 2 marks, implies M1
13	$n^3 + 4$ oe final answer and $\frac{11-n}{n+2}$ oe final answer and 2^{n+1} oe final answer	**6**	**B2** for $n^3 + 4$ oe final answer or **B1** for any cubic expression or for 3rd diff = 6 with at least two shown or for correct answer seen then spoilt **B2** for $\frac{11-n}{n+2}$ oe final answer or **B1** for fraction with numerator or denominator correct or for correct answer seen then spoilt **B2** for 2^{n+1} oe final answer or **B1** for answer of form $[a\times]2^{n[+b]}$ or $[a\times]\left(\frac{1}{2}\right)^{[b]-n}$ or for correct answer seen then spoilt	Condone use of other variable throughout Condone e.g. $n = n^3 + 4$ throughout B2 accept oe e.g. $\frac{10-(n-1)}{3+(n-1)}$ for 2 marks B2 oe e.g. $2(2^n)$, $4\times2^{n-1}$, $\left(\frac{1}{2}\right)^{-(1+n)}$ B1 for e.g. 4×2^n, $2\times2^{n-1}$, $\left(\frac{1}{2}\right)^{1-n}$ B0 for e.g. $\left(\frac{1}{2}\right)^{n+1}$, 2^{2n}, 4^n